Functional Connectivity Correlates to Individual Difference in Human Brains During Working Memory Task and Resting State

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Abstract

Working memory or short memory, the ability to hold information in mind during the ongoing flow of inputs, is one of the most important consecutive functions that enable human beings to achieve every type of daily life behavior. Working memory quality and capacity directly correlate to other cognitive behaviors such as attention, motor performance, long-term memory encoding and etc.. investigating the neural foundation behind it is not only important for understanding the core question of cognitive neuroscience -- how does the brain trigger behaviors but also beneficial for providing insights to understanding intelligence differences and even treatment for cognitively deficit diseases. The literature on brain and working memory has already identified brain areas such as the prefrontal cortex and basal ganglia including the thalamus and caudate that are involved in representing and holding information during a working memory task. In this study, we tested functional connectivity among human brains during both working memory task and rest states and how that is associated with individual differences in working memory performance. We identified that poor performance during working memory was represented by a flat and loosely connected brain network whereas during good performance in a working memory task, the brain, especially the cortex and thalamus was strongly connected. We found no difference between groups in the resting state network.

Introduction

The brain is a precise and complex network. Within only a few decades, the knowledge of the human brain expanded from a pure understanding of functional localization to being able to model and simulate the sophisticated neural interaction. Based on the large volumes of studies that confirmed that brain regions were dissociative for different functions, for example, brain regions that process visual information and motor function were separate, more recent studies succeed in building a connection between them. Both animal and human studies showed that the brain is a hierarchical structure that consists of independent and comprehensive systems yet the systems communicate in both structural and functional senses (Crossley, et.al., 2013; Shine, et.al., 2016; Smith, Abdala, Koizumi, Rybak and Paton, 2007).

Cognitive neuroscience studies that investigate how the brain as a dynamic network supports and regulates cognitive behaviors can tell an integrated story about the complexity of brains. And for high-level cognitive behaviors such as memory, understanding a single region is usually insufficient in providing full explanations for the neuronal mechanism behind the behavior (Laughlin and Sejnowski, 2003; Schedlbauer, Copara, Watrous and Ekstrom, 2014). In this project, we plan to utilize a network analysis approach and investigate how human brain network topology and dynamics associates with human working memory function as well as how brain regions connectivity differ between subjects represents the individual difference in working memory performance.

Working memory is one of the most important cognitive functions for both humans and animals to hold information for further processes in a short term. The working memory function tightly relates to and supports other functions such as long-term memory, attention, prospective processing and etc. (Engle and Kane,2000; Awh and Jonides, 2001; Engle and Kane,2000; Hertzog, Dixon, Hultsch and MacDonald, 2003). Due to technical limitations in human studies of neuroscience, most of the current results of how neuronal activation supports working memory stay at a region-scale level. Evidence showed that working memory was supported by multiple brain regions, including the prefrontal cortex, parietal cortex, posterior, and anterior cingulate cortex (Collette and Linden, 2002; Hampson, Driesen, Skudlarski, Gore and Constable, 2006). To understand how these regions work together in both a temporal dimension and weight dimension, researchers usually utilized a functional connectivity analysis. This type of tool usually uses graph theory to construct a time-series dynamic network on the neuroimaging data and provides a connection between distant areas (van den Heuvel and Hulshoff Pol, 2010). Current functional connectivity studies of working memory mainly focus on two direct -- how resting-state functional connectivity was impaired in individuals with abnormal working memory and how different types of information was processed by divergent brain networks within the working memory scope. A lack of studies asked whether the working memory network dynamics relate to normal individual performance differences. Also, while resting-state functional connectivity studies were prevalent in network analyses on the brain, we wondered how the brain network changes from rest to task state in relation to working memory capacity.

Here, we took published data by the Human Connectome Project, WU-Minn Consortium (Principal Investigators: David Van Essen and Kamil Ugurbil; 1U54MH091657) funded by the 16 NIH Institutes and Centers that support the NIH Blueprint for Neuroscience Research; and by the McDonnell Center for Systems Neuroscience at Washington University. The dataset included 100 human brains' functional magnetic resonance imaging during both resting state and a working memory task, along with behavioral and demographic data containing the task performance and individual information. In this project, we will run network analyses on the MRI data during both resting state and task state to try to answer the following questions: 1) can we construct the whole brain network mapping for human working memory and identify the hub regions which provide the most supports for working memory function. 2) can we identify network differences that contribute to individual differences in working memory performance? 3) how does the resting-state brain network differ in its topology from the task state brain network for working memory function. This is a reasonable question to ask for working memory because by its definition, working memory is a consistent state of the brain while awake. We wondered how the brain would change by adding more working memory load (giving a complex task rather than pure holding random information).

Method

Overall, the goal was to uncover the network community structure using hierarchical clustering. We aim to cluster functional connectivity and to map it to working memory task results. The main data we adopted is from resting-state fMRI scans as well as fMRI during the working

memory task. We preprocessed the imaging data with HCP minimal processing pipeline, which includes denoisation and normalization via ICA + FIX. To make parcellation, we used a multimodal parcellation (Glassier, et.al.,2016) suggested by HCP.

We focused on mapping both of the fMRI preprocessed data first. Several algorithms have been developed and successfully applied on large real networks. We used Louvain algorithm due to its compatibility in different null models. Our method is based on the paper from Akiki and Abdallah (2018) while the topic we focus on is different from their work. Their main purpose is to map to the individual's data while we focus on the overall phenomenon in specific working memory tasks and try to identify which brain hubs interact and support the achievement tasks successfully. The results will highly depend on the variation of the datasets whether we can distinguish specific regions which influence the performance of working memory. In order to avoid the bias relying on one specific indicator and more importantly, to solve the question of how brain dynamics correlates to individual differences in cognitive function ability, we plan to divide subjects into several groups to compare the performance in working memory under different conditions. Along the analyses, we might also add factors of demographic information, such as age, to investigate brain maps across different subjects. Through statistical analysis, we can summarize several critical regions for working memory.

Then we are also interested in running exactly the same experiments with task state data. We expect that the data under higher working load will be more complicated while mapping task state data to behavioral data is more intuitive. It will be a contribution if we can extract features with high correlation from task state data with resting state data as a benchmark. And through this analysis, we might also be able to discover time series changes of the network for different groups.

Since we are provided diffusion MRI data from the Human Connectome Project, which is usually for the research in structural connectivity. If we have spare time, we will explore the relationship between functional and structural connectivities. If the regions in a brain have a functional connection, then there might be an either directed or undirected structural connection. What's more attractive to us is to investigate the undirected paths which connect separate regions which support working memory simultaneously. We would like to answer the question of what intermediate regions build the physical connection for two-side regions even though those intermediate regions do not influence the working memory performance directly.

We also implement the Long Short-Term Memory (LSTM) model to answer if it's important to select critical features first when we predict the working memory performance with fMRI as the input data. LSTM is a specific recurrent neural network (RNN) architecture that was designed to deal with the vanishing gradient problem that may be encountered in conventional RNNs. It is known for addressing time series data. We built our model via keras with only a single hidden layer avoiding overfitting. During the experiment, we ran the same setting with 5 different seeds and 5-fold cross-validation. Because of the limited number of pages and the consideration about relevance of the course, the results and discussion of the LSTM model will be in Appendices.

Results

Behavioral Results

In order to analyze group-level functional connectivity, we firstly divided the subjects into groups based on the behavioral results. Shown in Fig. 1, the distribution of the accuracy skewed to the right with a mean of 86.28%. We divided the subjects into 2 groups -- poor performers and good performers using mean as the midline. This results in 40 good performers and 60 poor performers as comparison in following analyses.

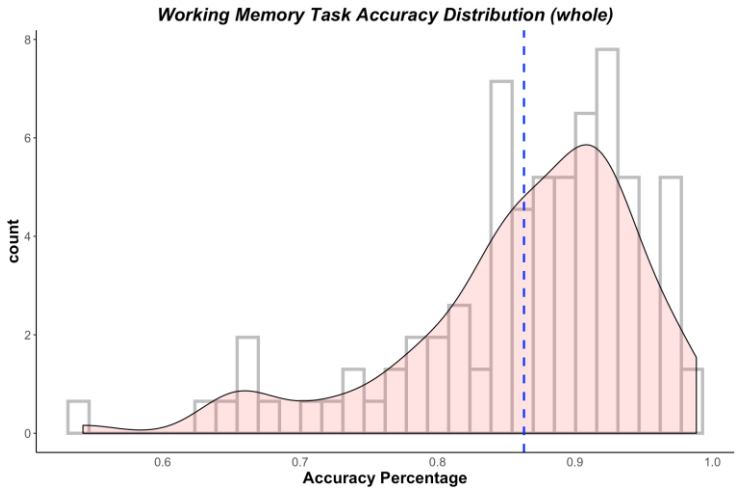

Figure 1 Working memory task performance summary across 100 subjects: this is the working memory task accuracy distribution plotted as bar histogram. The accuracy ranged from 56% to 98.8% with a median of 88.06% and mean of 86.28% (blue dash line). We splitted the subjects into two groups using the mean as mid-line.

Preprocessing

Following the preprocessing pipeline, we ended up with brain activity time-series for each subject, meaning we had time series activity for N locations of the brain across 405 time points during the working memory as well as the resting state. After parcellation, we got 21 regions in the brain, including both left and right cortex, thalamus, hippocampus, amygdala, putamen, caudate, cerebellum, ventral diencephalon and pallidum, plus brain stem.

Task State Network

Fig. 2 showed the correlation matrix for 21 regions' averaged activation across time-series within each group. We found a higher level of correlation distance in good performers than poor performers (deeper color shown in the right figure). Specifically, we found high correlation between cortex and thalamus as well as between cortex and caudate. After we ran hierarchical clustering, we found the optimized number of clusters to be 5 in both poor learners and good learners. We visualized this result in Fig.3, which represents the left hemisphere brain network during working memory task state. Corresponding to the correlation matrix results, we found stronger connections between nodes and a more activated network in good performers versus poor performers. Because we specifically targeted functional connectivity among the thalamus-

hippocampus-prefrontal cortex circuits, we also plotted out the brain activity across time series during the working memory task for these 3 areas in Fig.4. It showed that in good learners, the thalamus showed opposite activity direction from cortex while hippocampus showed similar trends to cortex. In the poor performers group, there's weak correspondence among these three areas.

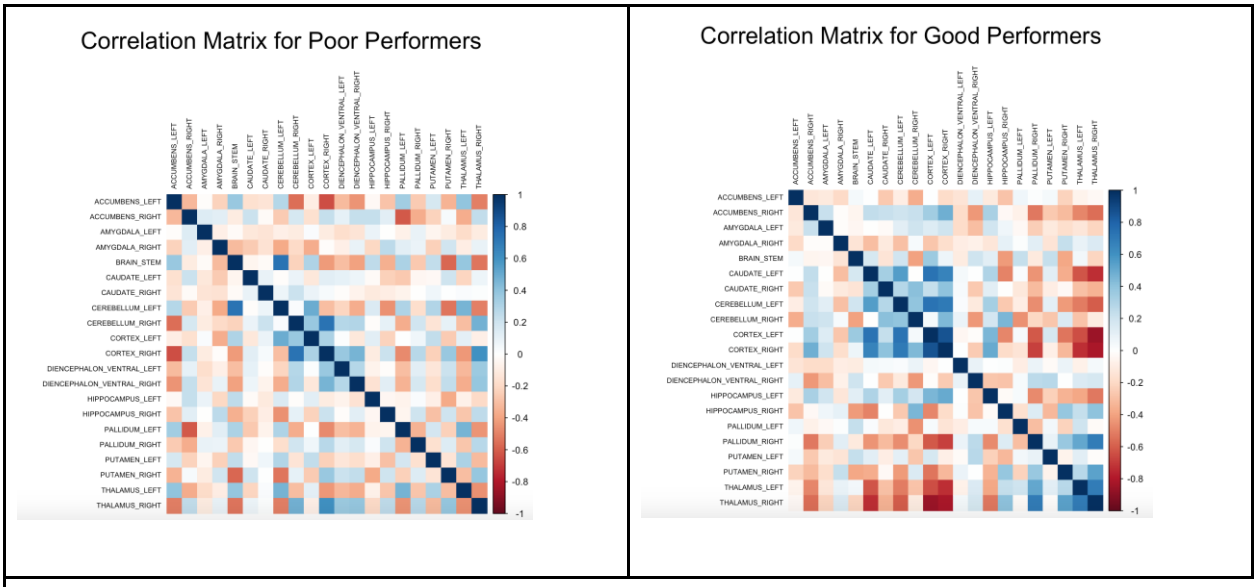

Figure 2 Correlation matrix for poor performers vs. good performers: after scaling, each brain region' time-series activity is averaged across subjects within the group. Pearson's correlation was calculated between all possible pairs. **Left**: poor performers. **Right**: good performers.

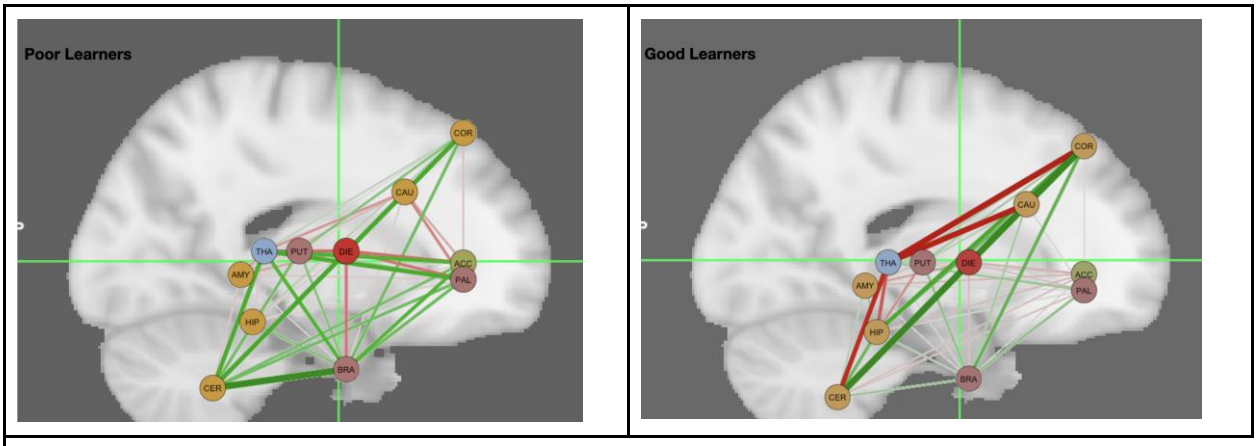

Figure 3 Brain's Left Hemisphere Network during Working Memory Task Across Subjects Compared Poor Performers vs. Good Performers: each node represents one parcellation of the left hemisphere. Nodes in the same color belong to the same cluster according to hierarchical clustering results. The edge represents correlation distance and negative correlations are shown in red and positive correlations are shown in green. The

Thickness and translucency represent the strength of the connection (absolute value of the correlation distance). **Left**: poor performers. **Right**: good performers.

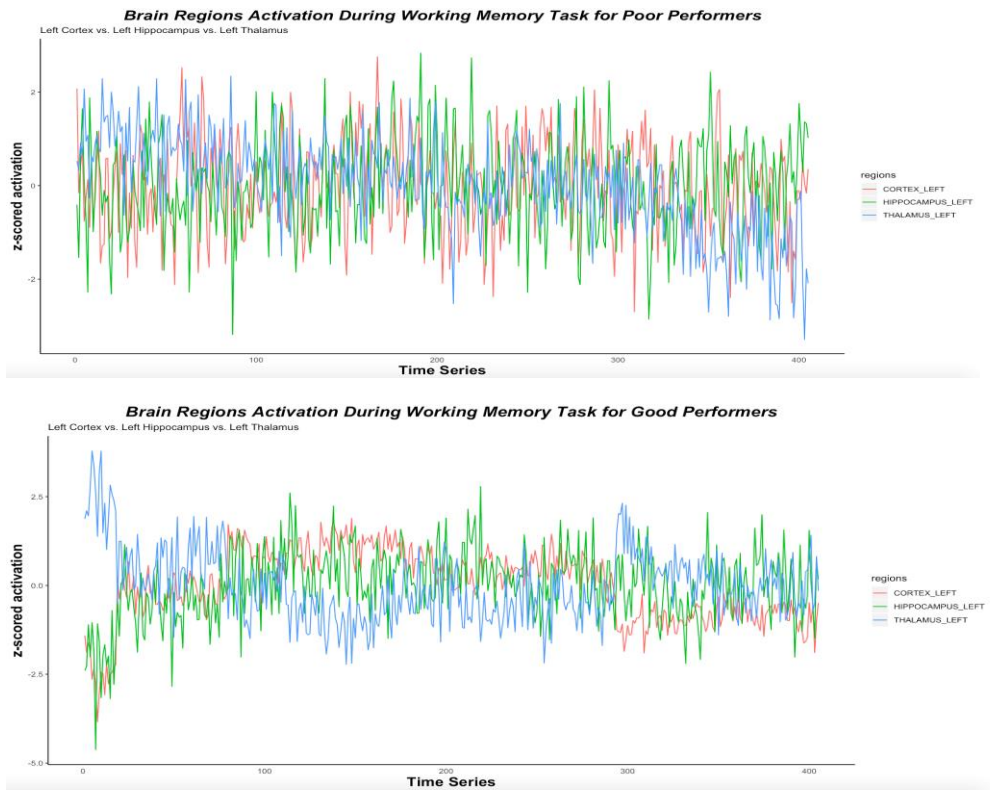

Figure 4 z-scored Neural Activation Across Time During Working Memory Task Comparing Left Thalamus with Left Cortex and Left Hippocampus. **Top**: poor performers. **Bottom**: good performers.

Resting State Network

Fig. 5 showed the correlation matrix for 21 regions' averaged activation across time-series within each group. We found that there is not a significant difference between good performers and bad performers. After we ran hierarchical clustering, we found that it's hard to optimize more meaningful clusters. We could only divide the regions in 2 clusters in both good learners and bad learners. However, thalamus, hippocampus, and prefrontal cortex circuit are at least in the same cluster. We visualized this result in Fig. 6, which represents the left hemisphere brain network during resting state. Corresponding to the correlation matrix results, we found the connections in good performers and poor performers are almost the same.

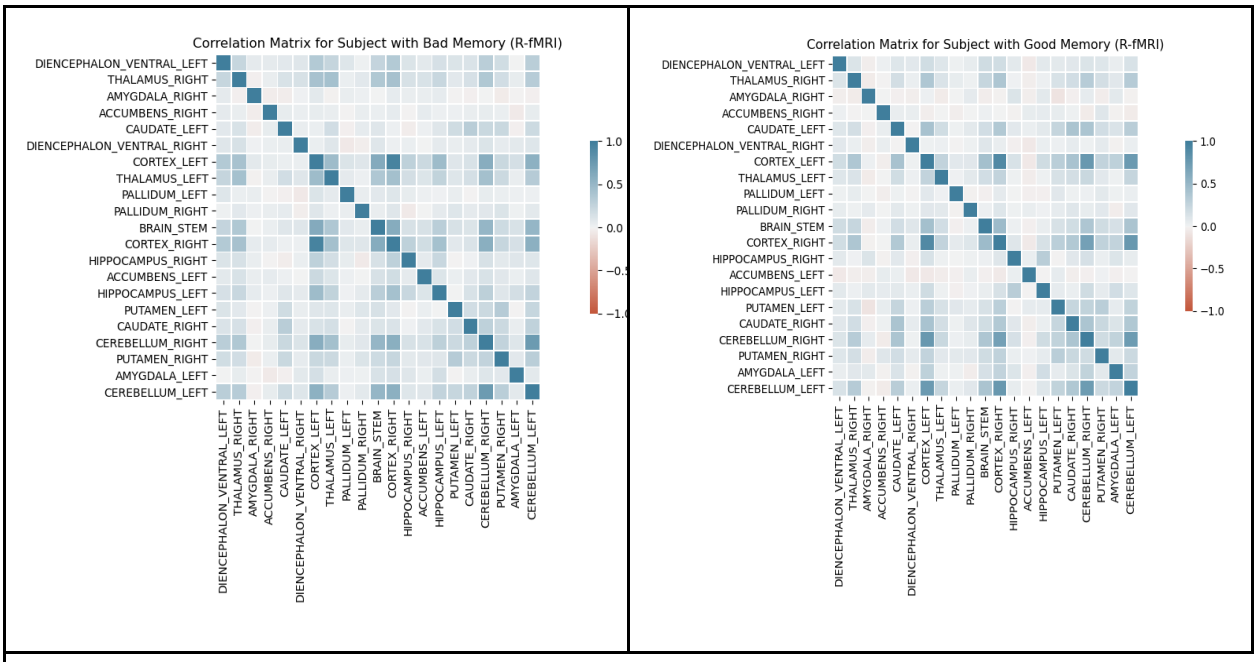

Figure 5 Correlation matrix for poor performers vs. good performers: after scaling, each brain region' time-series activity is averaged across subjects within the group. Pearson's correlation was calculated between all possible pairs. **Left**: poor performers. **Right**: good performers.

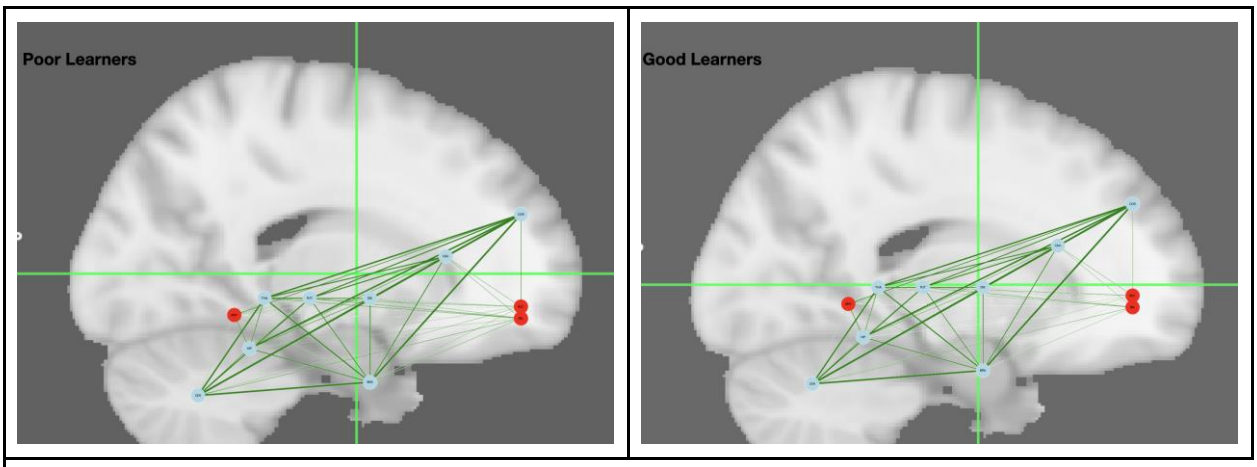

Figure 6 Brain's Left Hemisphere Network during Resting State Across Subjects Compared Poor Performers vs. Good Performers: each node represents one parcellation of the left hemisphere. Nodes in the same color belong to the same cluster according to hierarchical clustering results. The edge represents correlation distance and positive correlations are shown in green. The Thickness and translucency represent the strength of the connection (absolute value of the correlation distance). **Left**: poor performers. **Right**: good performers.

Discussion

In this project, we tried to solve an unclarified question -- why do humans show individual differences in their working memory capacities and how can we explain it from a brain connectivity dynamics point of view. By achieving so, we compared humans' brain network from two groups that differed a lot in their working memory performance. The results showed a much loosely connected brain or a weaker brain network when subjects did poorly in the working memory task. Meanwhile, for a good performance, we found a strong connection between cortex, thalamus and hippocampus. This circuit was found to be important for working memory in rats (Floresco, Braaksma and Phillips, 1999) and was emphasized in humans' long-term memory a lot (Ito et.al., 2015). This group comparison provided evidence for a direct connection between working memory capacities and the communications between these areas, indicating that the neural foundation for working memory is not supported by each brain part independently but an interconnected network. One future direction based on our results was to investigate whether performance differences within one subject could be explained by the same network behaviors found here. If so, this would provide new insights for train working memory deficit patients such as ADHD.

Unfortunately, we did not find a similar group difference in rest state. However, we found two main clusters in both groups during rest. We noticed that amygdala, caudate and putamen were less activated during rest. Considering these areas were shown to be essential for emotion or motor tasks, it is reasonable to observe a 'sleep' mode for them during rest.

Due to time limitation, we did not have a chance to investigate dynamics changes of the brain network. The dataset provided 405 time points and 1200 time points during task and rest. Thus, another future direction on this data is to divide the time series and depict functional connectivity's changes across the time. In doing so, it is possible to add a temporal component to the story neural explanation for humans' working memory skills.

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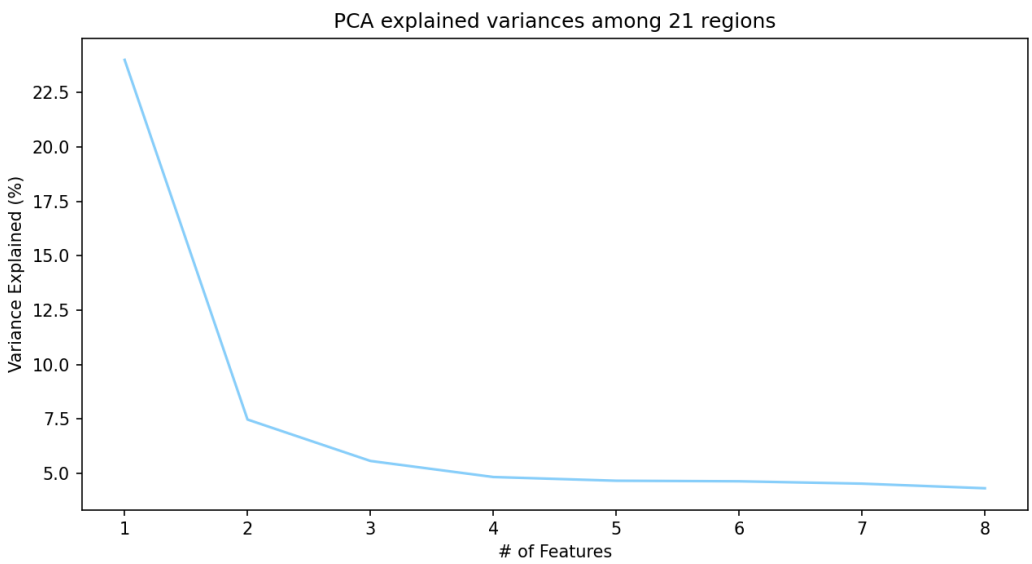
Appendices

LSTM Model for Resting fMRI

Since we could not evaluate the difference in functional connectivity between good performers and poor performers relying on correlation. We analyzed the data firstly by showing PCA in Fig. 9. We compared the variance explained between 21 regions and the 15 regions in the cluster with thalamus-hippocampus-prefrontal cortex circuits inside. Although the first feature from the bottom subplot in Fig. 9 is more than the one at the top subplot of Fig. 9, the gap could only indicate that we could get less variance with 15 regions than 21 regions. We still could not make sure that the data in 15 regions is more supportive of working memory tasks.

Table 1 showed the accuracy of prediction in working memory task performance. Although the difference is still not significant enough, we did observe the prediction of only considering the cluster (15 regions) which thalamus, hippocampus, and cortex are in is better than prediction with the whole regions (21 regions). In addition, during cross-validation, the fewer good memory subjects in the testing data, the more possible the selected data outperforms the whole data. We also listed other results using selected features via PCA from 15 regions.



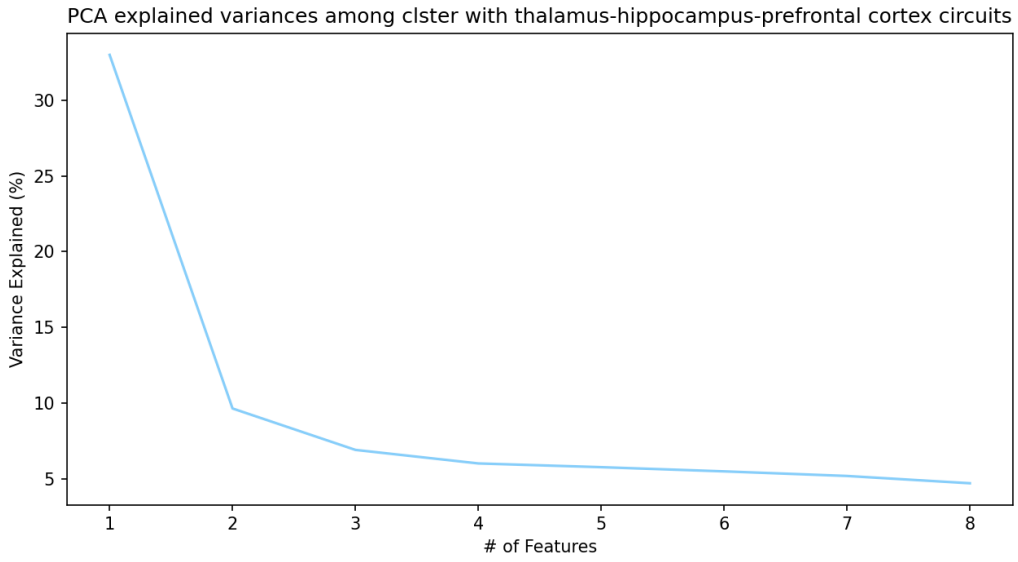


Figure 9 PCA in Resting fMRI: this is the PCA explained variance extracted as 8 features. **Top**: PCA for the whole 21 regions. **Bottom**: PCA for the clusters with thalamus-hippocampus-prefrontal cortex circuits inside.

21 regions	15 regions (cluster)	11 features in cluster	10 features in cluster	9 features in cluster	8 features in cluster	thalamus, hippocampus, cortex (left, right)
59.17%	59.69%	59.23%	58.02%	59.32%	58.46%	59.08%

Table 1. Accuracy of prediction of working memory tasks via resting fMRI